

proportionately.

Stage 3 (constant voltage): Constant current charge ends and the constant-voltage stage is invoked when the cell voltage reaches 4.2V. In order to maximise performance, the voltage regulation tolerance should be better than 1 per cent.

Stage 4 (charge termination): Unlike nickel-based batteries, it is not recommended to continue to trickle-charge Li-ion batteries. Continuing to trickle charge can cause plating of metallic lithium—a condition that makes the battery unstable. The result can be sudden, automatic and rapid disassembly.

Charging is typically terminated by one of two methods: minimum charge current or a timer (or a combination of the two). The minimum current approach monitors the charge current during the constant voltage stage and terminates the charge when the charge current diminishes in the range of 0.02C to 0.07C. The second method determines when the constant voltage stage is invoked. Charging continues for an additional two hours, and then the charge is terminated.

Charging in this manner replenishes a deeply depleted battery in roughly 2.5 to 3 hours.

Advanced chargers employ additional safety features. For example, the charge is suspended if the cell temperature is outside a specified window, typically 0°C to 45°C.

Li-ion charging: system considerations

A high-performance charging system is required to recharge any battery quickly and reliably. The following system parameters should be considered in order to ensure a reliable, cost-effective solution.

Input source. Many applications use very inexpensive wall cubes for the input supply. The output voltage is highly dependent on the AC input voltage and the load current being drawn from the wall cube.

In the US, the AC mains input volt-

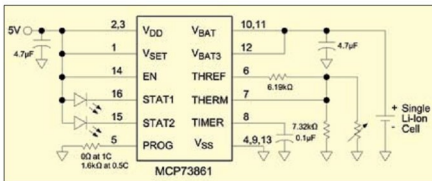


Fig. 4: Typical, fully integrated, linear solution

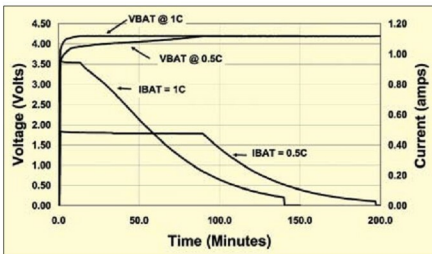


Fig. 5: MCP73861 charge cycle waveforms

age can vary from 90Vrms to 132Vrms for a standard wall outlet. Assuming a nominal input voltage of 120Vrms, the tolerance is +10 per cent, -25 per cent. The charger must provide proper regulation to the battery independent of its input voltage. The input voltage to the charger will scale in accordance to the AC mains voltage and the charge current:

$$V_o = \sqrt{2} \times V_{in} \times a - 1 \times (R_{EQ} + R_{PTC}) - 2 \times V_{FD}$$

R_{EQ} is the resistance of the secondary winding plus the reflected resistance of the primary winding (R_p/a^2), R_{PTC} is the resistance of the PTC and V_{FD} is the forward drop of the bridge rectifiers. In addition, transformer-core loss will slightly reduce the output voltage.

Applications that charge from a car adaptor can experience a similar problem. The output voltage of a car adaptor will be typically in the range

of 9V to 18V.

Constant-current charge rate and accuracy. The choice of topology for a given application may be determined by the desired constant current. Many high-constant-current, or multiple-cell, applications rely on a switch-mode charging solution for improved efficiency and less heat generation.

Linear solutions are desirable in low to moderate fast-charge-current applications for their superior size and cost considerations. However, a linear solution purposely dissipates excess power in the form of heat.

The tolerance on the constant-current charge becomes extremely important to a linear system. If the regulation tolerance is loose, pass transistors and other components will need to be oversized adding size and cost. In addition, if the constant-current charge is low, the complete